
Model 495 Gradient Former

Instruction Manual

**Catalog Number
165-4121**



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Section 1 Introduction

The Model 495 Gradient Former is primarily used to construct reproducible linear and exponential polyacrylamide gradient gels. It has a 100–1500 ml capacity, making it ideal for preparation of up to 12 identical large format gradient gels (both PROTEAN plus and PROTEAN II gels.)

The mixing (labeled light) and reservoir (labeled heavy) chambers have identical dimensions. The outlet from the mixing chamber leads into a multi-casting chamber. The exponential piston is available as an accessory; it is used to fix the volume in the mixing chamber to produce convex exponential gradients.

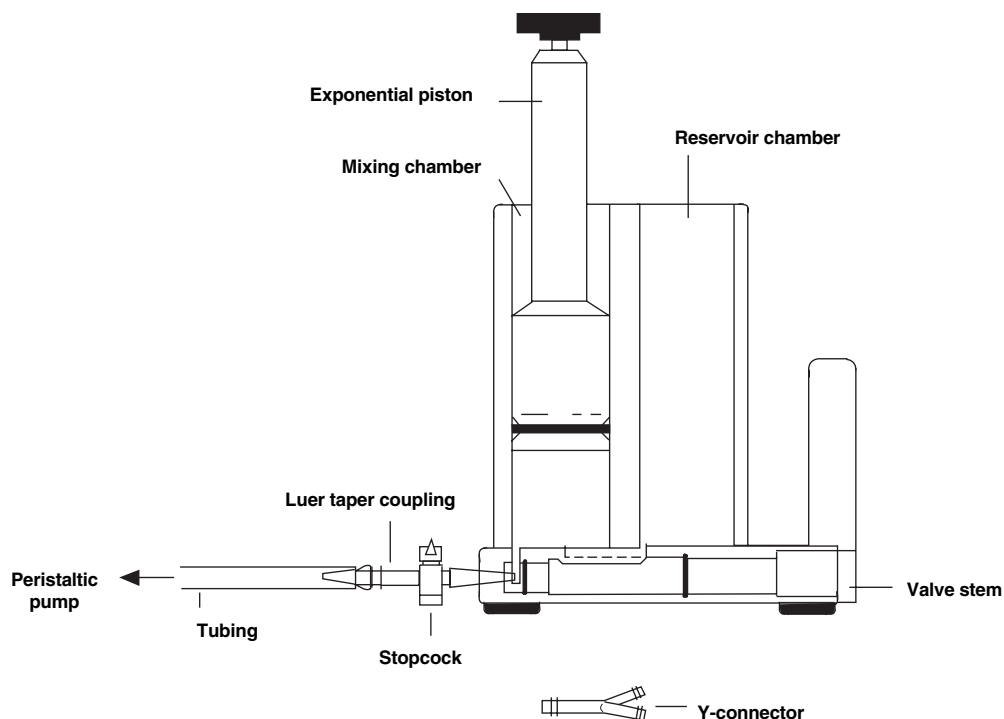


Fig. 1. Model 495 Gradient Former.

1.1 Specifications

Overall size	15.8 cm x 11.2 cm x 32.6 cm (W x L x H)
Weight	928 g
Capacity	100–1500 ml
Materials of construction	
Body	Machined from an acrylic block
Valve stem	Delrin®
Stopcock	Plastic
Piston	Delrin
Tubing	Tygon® 1/8" ID

Section 2

Pouring Multiple Gradient Gels in a Multi-Casting Chamber

Note: This procedure can be used to produce gels with either a linear or a convex exponential gradient.

Multi-Casting Chamber	Recommended number of gels
PROTEAN II Multi-Gel Casting Chamber	Cast at least 4 gels
PROTEAN plus Multi-Casting Chamber	Cast at least 4 gels

An inexperienced user should practice all steps ahead of time so that the procedure is completed quickly. The best guarantee of reproducibility from gradient to gradient is careful technique on the part of the operator.

Wear rubber gloves to prevent exposure to unpolymerized acrylamide, a neurotoxin.

2.1 Inlet Port Preparation for Casting From the Bottom

Connecting the Gradient Former to the Multi-Casting Chamber:

Cut a piece of Tygon tubing (3 mm I.D.) about 6–8 cm in length (a). Connect one end of the tubing to the multi-casting chamber port by twisting the tubing side to side until it is on about 1 cm. Fit a stopcock to the other end of the tubing (b). Attach the luer connector to the stopcock (c). Tubing from the gradient former attaches to the luer connector (d). The other end of the gradient former tubing then is fitted with a luer connector (e), and this fits into a stopcock (f). This stopcock attaches directly to the gradient former. Close all valves to prepare for casting (Section 2.2).

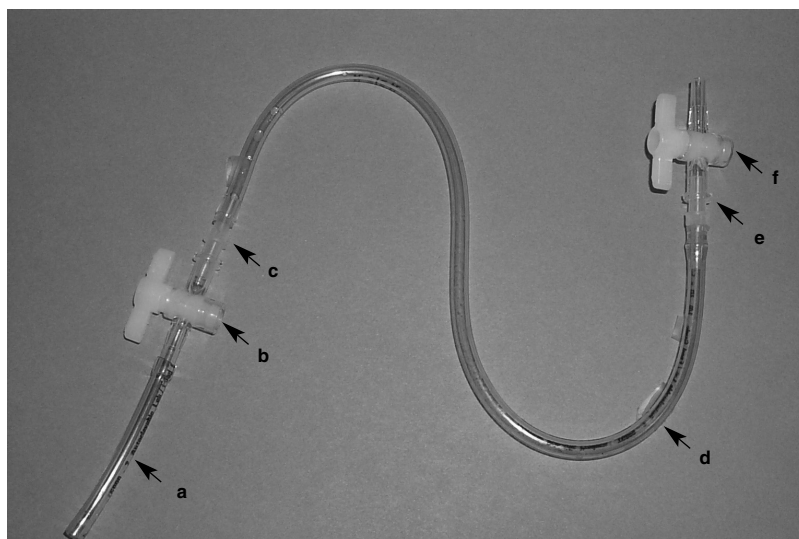


Fig. 2. Tubing and fittings to connect the Multi-Casting Chamber to the Gradient Former.

2.2 Casting Gels

The first time gels of a certain thickness are cast, it is necessary to empirically determine the required volume of acrylamide. If you are using the PROTEAN plus multi-casting chamber, assemble the stack of glass plates (as described in the multi-casting chamber manual) and inject a measured volume of water through the stopcock. Prepare this volume

(+ 10 ml) of acrylamide (see sample calculation below). Disassemble the chamber and dry all components.

Note: Wear rubber gloves while performing the following procedure to prevent accidental exposure to unpolymerized acrylamide, a neurotoxin.

2.2.1 Casting LINEAR Gradient Gels

To create a linear gradient, the volume in each chamber is generally $\frac{1}{2}$ the total gel volume required.

Note: GRADIENT GELS ONLY—the size of the groove has been optimized so that the monomer solution enters all of the hinged spacer plates at the same time, rather than one at a time (wicking). However, the larger dimension increases the amount of monomer solution required to fill the V-groove, thus slightly altering the actual gradient formed in the gels. With this in mind, TWO solutions are recommended:

- Adjust the heavy solution by approximately 0.5% to help compensate for the V-groove. For example: If 8–16% linear gradient gels are being cast, prepare 16.5% acrylamide/Bis for the heavy solution (see sample calculation below).

- Adjust the amount of light and heavy solution so that they are not equal.

For example: If the total volume required to cast twelve 4–20% gradient gels is 1100 ml, prepare 555ml of the light monomer solution (4%) and 545 ml of the heavy monomer solution (20%). (See sample calculation and technical directions below.)

1. Place the gradient former on a magnetic stir plate and add a stir bar to the mixing chamber labeled "light". (See section 2.1 for inlet port preparation.) Place the gradient former above the top of the plate set, but minimize the length of tubing between it and the multi-casting chamber. Be certain that both stopcocks are closed. The valve stem lever should be turned to the 45°/closed position.

Note: The level of the gradient former stopcock must be placed above the top of the gel sandwich. This creates a hydrostatic head large enough to cast the gels within 10 minutes from the time the initiators are added to the light solution. To create uniform gradients, all of the acrylamide must be in the gel sandwich before polymerization begins. See Figure 3 for proper set-up.

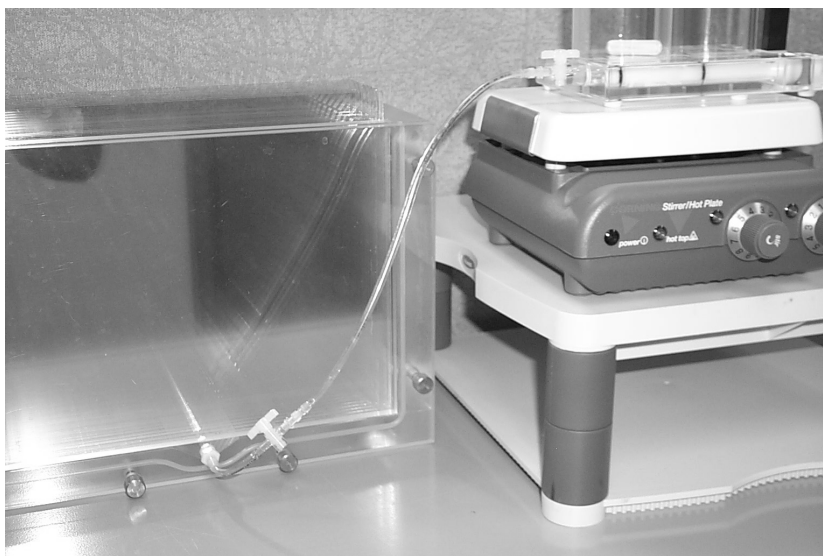


Fig. 3. Proper set-up to cast 12 gradient gels.

2. Prepare the heavy and light monomer solutions.

Note: For the larger format gels (≥ 20 cm in width) it is necessary to **decrease the TEMED** concentration by 50% (for a final concentration of **0.025%**). This will extend the polymerization time to approximately 20 minutes, providing time to complete casting and for the monomer solution to settle properly. *Also, adding **10% GLYCEROL** to the HEAVY solution is important when preparing GRADIENT GELS. Glycerol adds "weight" to the heavy solution, the solution that flows into the multi-casting chamber behind the light solution. This added weight prevents the heavy solution from pushing up through the middle into the light solution. (This would cause a ripple in the gradient.) Glycerol is washed out during regular staining and protein digestion procedures that use acetonitrile and ammonium bicarbonate. It will not interfere with protein identification (using mass spectrometry).*

Sample Calculation: Linear Gradient Gels

Objective: Cast 12 gels, 2.0 mm, 25 x 20.5 cm, 8–16% acrylamide.

Place 12 clean hinged spacer plates and separation sheets in the multi-casting chamber (as described in the multi-casting chamber instruction manual) and measure the volume required to fill the plates with water. In this example, the volume required is 1330 ml. Disassemble and dry all components. Prepare 1340 ml of solution, 675 ml of light monomer solution (8%) and 665 ml of heavy monomer solution (16.5%). Use the standard equation, $C_1V_1 = C_2V_2$.

To prepare 675 ml (8%)

(8%) (675 ml) = (X ml) (30% acrylamide/Bis Stock)	180 ml
(0.375M Tris-Cl)(675 ml) = (X ml) (1.5M Tris-Cl pH 8.8 Stock)	169 ml
(675 ml) – (180 ml + 169 ml) = ml Water	326 ml
(0.05% APS) (675 ml) = (X ml) (10% APS Stock)	3.375 ml
(0.025% TEMED) (675 ml) = (X ml) (100% TEMED Stock)	169 μ l

To prepare 665 ml (16.5%)

(16.5%) (665 ml) = (X ml) (30% acrylamide/Bis Stock)	366 ml
(0.375M Tris-Cl)(665 ml) = (X ml) (1.5M Tris-Cl pH 8.8 Stock)	166 ml
(10% GLYCEROL) (665 ml) = (X ml) (100% GLYCEROL)	66.5 ml
(665 ml) – (366 ml + 166 ml + 66.5 ml) = ml Water	66.5 ml
(0.05% APS) (665 ml) = (X ml) (10% APS Stock)	3.325 ml
(0.025% TEMED) (665 ml) = (X ml) (100% TEMED Stock)	166 μ l

3. Combine all reagents except the initiators (usually APS and TEMED) and degas the solutions for 15 minutes under a vacuum. Degassing the solutions removes oxygen (oxygen inhibits polymerization.)
4. Immediately prior to pouring, add TEMED and APS to both solutions, mix gently, and pour the appropriate monomer solutions into the gradient chambers. The light solution (the one with the lower acrylamide concentration) should be placed in the mixing chamber labeled "light", and the heavy solution in the reservoir chamber labeled "heavy". The level of the solutions in the two chambers will NOT be equal at this time.
5. Turn on the stirring bar in the mixing chamber to a steady speed and maintain this same speed throughout casting.

Note: If flow rate is kept constant, gradients will be reproducible. Tubing size, the volume of acrylamide in the chambers, and the *rate of stirring* must be kept constant. If the rate of stirring changes, the vortex that is created will also vary. The speed controls the amount of acrylamide pulled into the mixing (light) chamber from the reservoir (heavy) chamber when the connection between the two chambers is opened. Variation in stirring can disrupt linear gradient formation. Also, the vortex should be off-center inside the mixing chamber. This can be achieved by moving the gradient former on the stir plate to offset the stir bar.

6. Start casting the gels by opening the STOPCOCK valve to the multi-casting chamber and the STOPCOCK valve to the gradient former. *The valve stem lever on the gradient former is closed at this time.* Allow the light monomer solution to enter the multi-casting chamber until the level of solution in the mixing chamber is EQUAL to the level of the heavy monomer solution in the reservoir chamber. The light solution should be close to entering the plates in the casting chamber. This should help compensate for the V-groove in the multi-casting chamber.

Note: The gradient must be poured as quickly as possible, without mixing the gradient solution in the casting chamber. In addition to decreasing the amount of initiators, a peristaltic pump can be used to pump the entire set of gradients within 10 minutes. However, if it is still not possible to complete the operation in 10 minutes with a pump, decrease the concentration of initiators to slow polymerization. An inexperienced user should practice all steps ahead of time so that the procedure of starting the gradient is completed quickly.

Note: The larger format gels ($\geq 20\text{cm}$ in width) require a decrease in the TEMED concentration by 50% (for a final concentration of 0.025%).

7. Open the VALVE STEM lever on the gradient former once the levels are EQUAL in the two chambers, and begin mixing the solutions and creating the gradient. DO NOT allow any air bubbles to enter the plates.
8. After casting the gels to the top of the plates, CLOSE the stopcock on the multi-casting chamber and on the gradient former. Carefully remove the luer connector from the stopcock connected to the multi-casting chamber. Put the tubing into a beaker and drain the excess acrylamide from the gradient former. Insert the combs at an angle; this will help minimize bubble formation in the wells.

Caution: Immediately flush the gradient former and tubing with water to prevent polymerization of residual acrylamide within the gradient former.

2.2.2 Casting CONVEX Exponential Gradient Gels

For a convex exponential gradient, the volume of the light solution in the mixing chamber is equal to $\frac{1}{4}$ the total gel volume and the heavy solution in the reservoir chamber will be equal to the total gel volume. Creating a convex gradient gel will "open up" the bottom region of a gel. Because the volume of the mixing chamber is fixed, not all of the gel solution will be delivered to the gel sandwich. Discard any remaining solution.

Note: The maximum volume of each chamber is 750 ml.

Note: In this case, the volume of monomer held in the V-shaped groove does not affect the quality of the convex gradient.

- a. Place the gradient former on a magnetic stir plate and add a stir bar to the chamber labeled "light". (See Section 2.1 for inlet port preparation.) Place the gradient former above the top of the plate set, but minimize the length of tubing between it and the multi-casting chamber.

Note: The level of the gradient former stopcock must be placed above the top of the gel sandwich. This creates a hydrostatic head large enough to cast the gels within 10 minutes from the time the initiators are added to the light solution. To create uniform gradients, all of the acrylamide must be in the gel sandwich before polymerization begins. See Figure 3 for proper set-up.

b. Prepare the heavy and light monomer solutions.

Note: For the larger format gels (≥ 20 cm in width) it is necessary to **decrease the TEMED** concentration by 50% (to a final concentration of **0.025%**). This will extend the polymerization time to approximately 20 minutes, providing time to complete casting and for the monomer solution to settle properly. *Also, adding 10% GLYCEROL to the HEAVY solution is important when preparing GRADIENT GELS. Glycerol adds "weight" to the heavy solution, the solution that flows into the multi-casting chamber behind the light solution. This added weight prevents the heavy solution from pushing up through the middle into the light solution. (This would cause a ripple in the gradient.) Glycerol is washed out during regular staining and protein digestion procedures that use acetonitrile and ammonium bicarbonate. It will not interfere with protein identification (using mass spectrometry).*

Sample Calculation: Convex Gradient Gels

Objective: Cast 6 gels, 2.0 mm thick, 25 x 20.5 cm, 8–18% convex gradient.

Note: Only 6 convex gradient gels can be prepared at a time, since the maximum volume of each cylinder is 750 ml.

Place 6 clean gel cassettes and separation sheets in the multi-casting chamber (as described in the multi-casting chamber manual) and measure the volume required to fill the cassettes with water. For this example, the volume required is 720 ml.

Disassemble and dry all components. Prepare 720 ml of solution, 180 ml of light monomer solution (8%) and 720 ml of heavy monomer solution (18%). Use the standard equation, $C_1 V_1 = C_2 V_2$.

To prepare 180 ml (8%)

(8%) (180 ml) = (X ml) (30% acrylamide/Bis Stock)	48 ml
(0.375M Tris-Cl)(180 ml) = (X ml) (1.5M Tris-Cl pH 8.8 Stock)	45 ml
(180 ml) – (48 ml + 45 ml) = ml Water	87 ml
(0.05% APS) (180 ml) = (X ml) (10% APS Stock)	900 μ l
(0.025% TEMED) (180 ml) = (X ml) (100% TEMED Stock)	45 μ l

To prepare 720 ml (18%)

(18%) (720 ml) = (X ml) (30% acrylamide/Bis Stock)	432 ml
(0.375M Tris-Cl)(720 ml) = (X ml) (1.5M Tris-Cl pH 8.8 Stock)	180 ml
(10% GLYCEROL) (720 ml) = (X ml) (100% GLYCEROL)	72ml
(720 ml) – (432 ml + 180 ml + 72 ml) = ml Water	36 ml
(0.05% APS) (720 ml) = (X ml) (10% APS Stock)	3.6 ml
(0.025% TEMED) (720 ml) = (X ml) (100% TEMED Stock)	180 μ l

- Combine all reagents except the initiators (usually APS and TEMED) and degas the solutions for 15 minutes under a vacuum. Degassing the solutions removes the oxygen, which inhibits polymerization.
- Immediately prior to casting, add TEMED and APS to both solutions, mix gently, and pour the light monomer solution (the one with the lower acrylamide concentration) into the mixing chamber (labeled light) and the heavy monomer solution into the reservoir chamber (labeled heavy).

- e. Fix the light solution volume with the exponential piston by inserting the piston into the mixing chamber to 1 cm above the level of the monomer and tightening the screw-top handle to hold the piston in place.
- f. Turn on the stirring bar in the mixing chamber to a steady speed and maintain this same speed throughout casting.

Note: If the flow rate is kept constant, the gradients will be reproducible. Tubing size, the volume of acrylamide in the chambers, and the rate of stirring must be kept constant. Also, the vortex should be off-centered within the mixing chamber. This can be achieved by moving the gradient former on the stir plate and simply offsetting the stir bar in the mixing chamber.

- g. Start casting the gels. First make sure the stopcock valve to the multi-casting chamber is open. Then at the same time open the STOPCOCK valve to the gradient former and turn the VALVE STEM LEVER to its vertical/open position. As the acrylamide from the mixing chamber is removed the reservoir solution is drawn into the mixing chamber. The mixing chamber solution appears to be stationary. DO NOT allow any air bubbles to enter the gel cassettes.
- h. After casting the gels to the top of the gel cassettes, CLOSE the stopcock on the multi-casting chamber and on the gradient former. Carefully remove the luer connector from the stopcock connected to the multi-casting chamber. Put the tubing into a beaker and drain the excess acrylamide from the gradient former. For a continuous buffer system, now insert the combs at an angle, this will help minimize bubble formation in the wells. Otherwise, overlay each gel **as quickly as possible** with 1.0–2.0 ml of water or water-saturated isobutanol. Each gel **must** have the same volume of overlay solution.

Caution: Immediately flush the gradient former and tubing with water to prevent polymerization of residual acrylamide within the gradient former.

2.2.3 Casting CONCAVE Exponential Gradient Gels

The Multi-Casting Chamber can not be used to cast concave exponential gradient gels because it is intended for bottom filling, and concave gradient gels are cast from the top. Therefore only PROTEAN II xi and XL concave gradient gels (≥ 20 cm in length) can be prepared ONE AT A TIME in the slab gel casting stand (catalog number 165-1911).

Note: The Model 495 Gradient Former has a minimum volume of 100 ml therefore the Model 485 Gradient Former (catalog number 165-4120) is required to cast concave exponential gradient gels (refer to the Model 485 Gradient Former manual for operating instructions).

Section 3

Theory of Linear and Exponential Gradient Gels

3.1 Linear Gradients

Linear gradients are described by the equation:

$$C_t = C_M + (C_R - C_M)V_t/2V_0$$

where C_t = concentration of gradient being delivered at any time

C_M = initial concentration in the mixing chamber

C_R = concentration in reservoir

V_t = volume delivered at time t

V_0 = original volume in each chamber

A typical linear gradient is shown in Figure 4. (Insert Figure 7.1 from M1652000) The slope of the gradient is changed by changing the acrylamide concentration in the mixing and/or reservoir chamber.

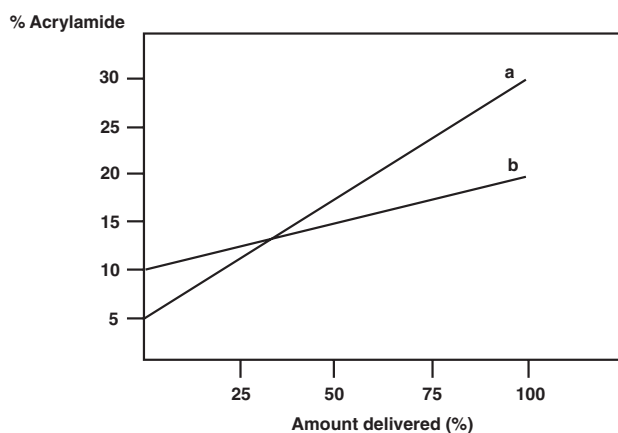


Fig. 4. Linear gradient gel. a. 5–30% gradient. b. 10–20% gradient.

3.2 Exponential Gradients

Both convex and concave exponential gradients are described by the equation

$$C_t = C_R - (C_R - C_M)e^{-V_t/V_M}$$

where e = natural base (2.718), and

V_M = volume of acrylamide in mixing chamber; the other symbols are as above.

Figure 5 shows how the % acrylamide varies with the amount of acrylamide delivered.

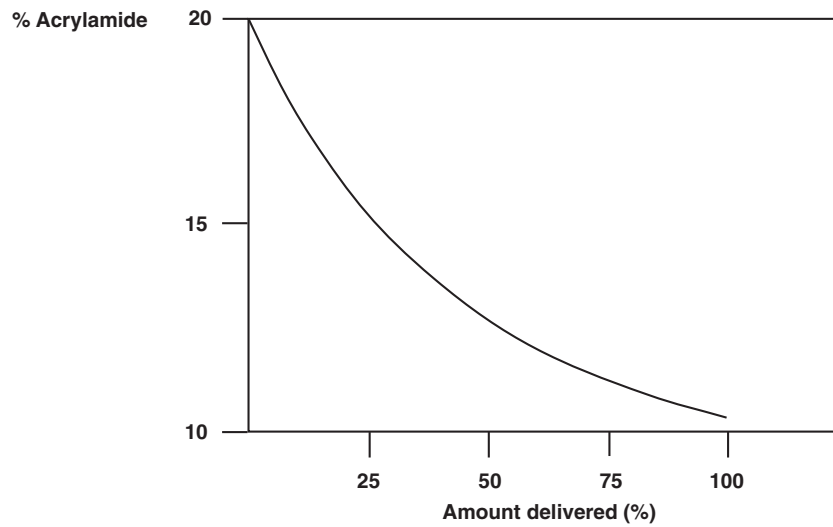


Fig. 5. Concave gradient gel.

Section 4 Protocols

Preparation of Stock Solutions for SDS-PAGE Slab Gels (Laemmli¹ buffer system)

Note: See Section 7 for available premixed buffers

- A. Acrylamide/Bis (30% T, 2.67% C)
Acrylamide 146.0 g
N'N'-BIS-methylene-acrylamide 4.0 g
Add distilled water to 500 ml. Filter and store at 4 °C in the dark. Maximum shelf life under these conditions is 30 days.
- B. 1.5M Tris-HCl, pH 8.8
54.45 g Tris base
150 ml distilled water
Adjust to pH 8.8 with 1 N HCl. Add distilled water to 300 ml. Store at 4 °C.
- C. 0.5 M Tris-HCl, pH 6.8
6 g Tris base
60 ml distilled water
Adjust to pH 6.8 with 1 N HCl. Add distilled water to 100 ml. Store at 4 °C.
- D. 10% (w/v) SDS
Dissolve 10 g SDS in water with gentle stirring. Add distilled water to 100 ml.

E. 10% ammonium persulfate (w/v)

100 mg ammonium persulfate into 1 ml of distilled water. Make fresh daily.

F. Sample buffer

Distilled water	4.0 ml
0.5 M Tris-HCl, pH 6.8	1.0 ml
Glycerol	0.8 ml
10% SDS	1.6 ml
β- mercaptoethanol	0.4 ml
0.05% (w/v) bromophenol blue (in water)	<u>0.2 ml</u>
	8.0 ml

Dilute the sample at least 1:4 with sample buffer. Heat at 95 °C for 4 minutes.

G. 5x Running Buffer

Tris base	45.0 g
Glycine	216.0 g
SDS	15 g

Distilled water in 3 L. Do not adjust pH with acid or base. Store at 4 °C. Warm to 37 °C before use if precipitation occurs. Dilute 300 ml of 5x stock with 1.2 l distilled water to make 1.5 l 1x buffer.

¹ Laemmli, U.K., *Nature*, **227**, 680 (1970).

Section 5

Gradient Former Care and Maintenance

After use, disassemble the gradient former and rinse all parts with distilled water. A Tubing Connection Kit (catalog number 165-2008) includes replacement Tygon tubing, a luer fitting, stopcock, and a Y connector. Do not use organic solvents, strong acid solutions or ethanol in the gradient former.

Section 6

Product Information

Catalog Number	Description
165-4121	Model 495 Gradient Former
165-2005	Exponential Piston for Model 495 Gradient Former
165-2008	Tubing Connection Kit , includes stopcock, tapered luer coupling, tubing (1/8 inch ID, 3 feet), and Y-connector

Section 7

Reagents for Electrophoresis

Catalog Number	Description
Premixed Electrophoresis Buffers	
161-0732	10x Tris/Glycine/SDS Premixed Buffer , 1 L
161-0733	10x Tris/Boric Acid/EDTA Premixed Buffer , 1 L
161-0734	10x Tris/Glycine Premixed Buffer , 1 L
161-0741	10x TBE Extended Range Premixed Buffer , 1 L
161-0743	50x Tris/Acetic Acid/EDTA Premixed Buffer , 1 L
161-0744	10x Tris/Tricine/SDS Premixed Buffer , 1 L
161-0772	10x Tris/Glycine/SDS Premixed Buffer , 5 L
161-0770	10x Tris/Boric Acid/EDTA Premixed Buffer , 5 L
161-0771	10x Tris/Glycine Premixed Buffer , 5 L
161-0758	10x TBE Extended Range Premixed Buffer , 6 x 1 L
161-0773	50x Tris/Acetic Acid/EDTA Premixed Buffer , 5 L
161-0760	10x Tris/Tricine/SDS Premixed Buffer , 6 x 1 L
Premixed Sample Buffers	
161-0737	Laemmli Sample Buffer , 30 ml
161-0738	Native Sample Buffer , 30 ml; (Store at 2–8 °C.)
161-0739	Tricine Sample Buffer , 30 ml
161-0767	Nucleic Acid Sample Buffer , 5X, 10 ml
161-0768	TBE-Urea Sample Buffer , 30 ml
Acrylamide Solutions	
161-0158	30% Acrylamide/Bis Solution 37.5:1 , 500 ml
161-0140	40% Acrylamide Solution , 500 ml
161-0142	2% Bis Solution Crosslinker , 500 ml
161-0144	40% Acrylamide/Bis Solution 19:1 , 500 ml
161-0146	40% Acrylamide/Bis Solution 29:1 , 500 ml
161-0148	40% Acrylamide/Bis Solution 37.5:1 , 500 ml
161-0154	30% Acrylamide/Bis Solution 19:1 , 500 ml
161-0156	30% Acrylamide/Bis Solution 29:1 , 500 ml

Catalog Number	Description
Premixed Acrylamide I Bis Powders	
161-0123	Acrylamide/Bis 19:1 , 150 g
161-0124	Acrylamide/Bis 29:1 , 150 g
161-0125	Acrylamide/Bis 37.5:1 , 150 g
Premixed Gel Casting Buffers	
161-0798	Resolving Gel Buffer 1.5 M Tris-HCl , pH 8.8, 1L
161-0799	Stacking Gel Buffer 0.5 M Tris-HCl , pH 6.8, 1 L



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